

Assignment 7

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1 Setup

1.1 Objective

1. Perform Molecular Dynamic (MD) simulation of interacting many bodies system.

1.2 Model

Interacting potential $U(r_{jk})$ is

$$U(r_{jk}) = \epsilon \left(\frac{a_{jk}}{r_{jk}} \right)^{12} + C_{jk} \quad (r_{jk} < a_{\text{cut}}) \quad (1)$$

Potential energy U and kinetic energy K is

$$U = \sum_{i < j}^N U(r_{ij}) \quad (2)$$

$$K = \sum_{i=1}^N \frac{1}{2} m v_i^2 \quad (3)$$

It's also useful to consider average energy per particle.

$$u = U/N \quad \kappa = K/N \quad (4)$$

Using the following non-dimensionalize parameter.

$$t = t_0 \tilde{t} \quad (5)$$

$$x = a \tilde{x} \quad (6)$$

$$v = \frac{a}{t_0} \tilde{v} \quad (7)$$

$$T^* = k_B T / \epsilon \quad (8)$$

where

$$t_0 = \sqrt{m a^2 / \epsilon} \quad (9)$$

1.3 Theory

1.3.1 Maxwell-Boltzmann distribution

Maxwell-Boltzmann distribution in 2D is given by $f(v)$

$$f(v) = \frac{m}{k_B T} v \exp\left(-\frac{mv^2}{2k_B T}\right) \quad (10)$$

1.3.2 Equipartition theorem

A system with f degrees of freedom under thermal equilibrium at temperature T will have average energy per particle $\langle e \rangle$

$$\langle e \rangle = \frac{f}{2} k_B T \quad (11)$$

Namely, in the case of 2D. Average kinetic energy will be

$$\langle \kappa \rangle = k_B T \quad (12)$$

1.3.3 Virial-Theorem

If the potential U is central and propotional to power n of distance r

$$U \propto r^n \quad (13)$$

Then the time average $\langle \dots \rangle$ of kinetic energy K and potential energy U have the following relation.

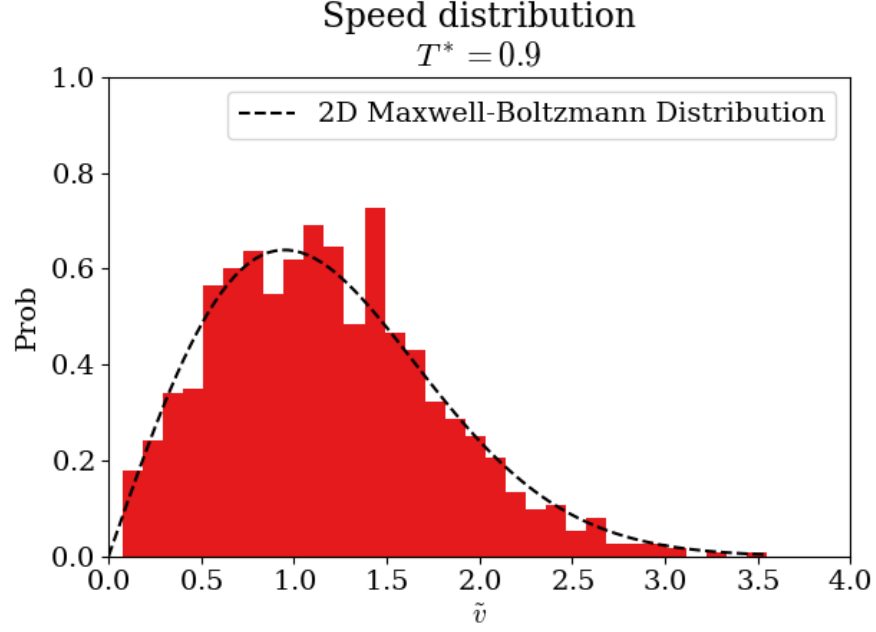
$$\langle K \rangle = \frac{n}{2} \langle U \rangle \quad (14)$$

1.4 Method

1. Use Langevin heat bath to prepare particles in thermal equilibrium state at dimensionless temperature $T^* = k_B T / \epsilon = 0.9$
2. Use positions and velocities in 1. as initial conditions and perform MD simulation where the force is only from interaction 1

2 Result & Discussion

First let's look at the state in thermal equilibrium with Langevin heat bath at temperature $T^* = 0.9$



⊠ 1: speed distribution

The histogram of speed v aligns very well with Maxwell-Boltzmann Distribution. Despite the starting Langevin equation is merely stochastic differential equations with constrain such that the long time limit of v^2 will approach $2k_B T$, the steady state distribution of v is Maxwell-Boltzmann.

This poses an interesting question. Why specifying the long term limit, make the random variable v follows MB distribution? Can we analytically derive the probability distribution? Is there any another SDE that results with the same probability distribution in steady state?

Next, we consider the potential and kinetic energy of the system.

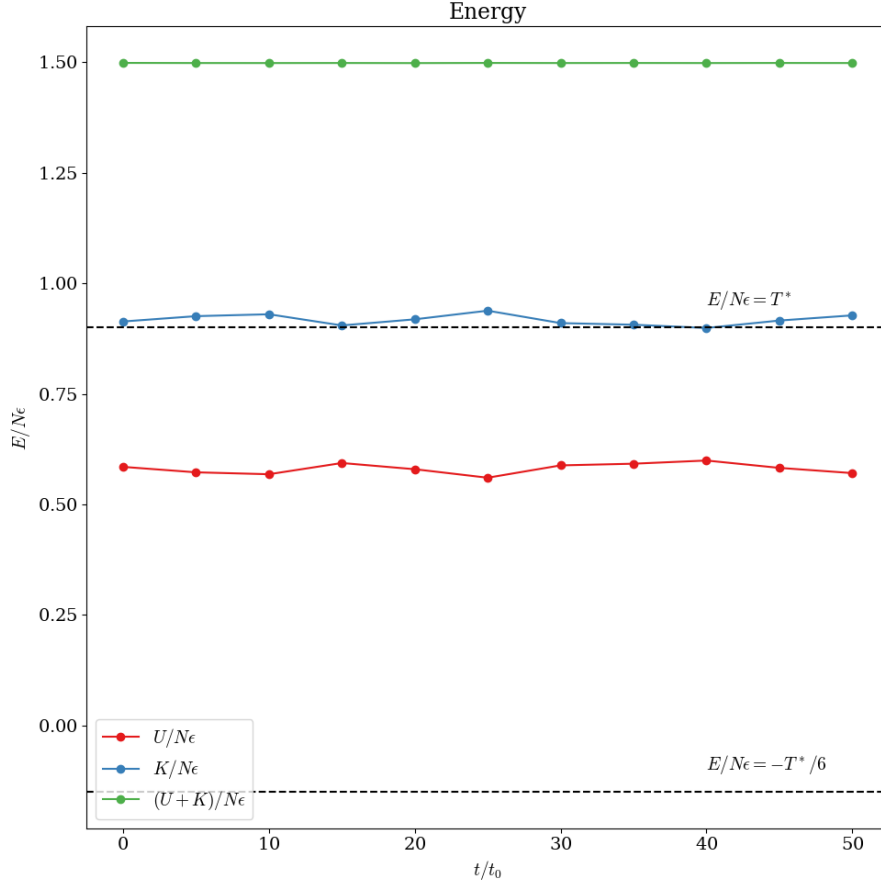


Figure 2: Energy

As we can see, the kinetic energy K and potential energy U fluctuate but the total mechanical energy E remains constant all the time.

The average kinetic energy aligns very well with Equipartition theorem, fluctuating around non-dimensionalized energy T^* . However, the average potential energy can not be described by Virial theorem despite the fact that LJ potential $\propto r^{-12}$. Numerical analysis showed that

$$U/N\epsilon \sim 0.6 = \frac{2}{3} K/N\epsilon \quad (15)$$

meaning that instead of $n = 3$, LJ potential satisfies Virial theorem as $n = 3$. This discrepancy is worth to investigate. Some possible explanations are, 1. using wrong statement of Virial theorem. 2. the use of cutoff distance.